

HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE

Software Engineer — HCL Technologies, Chennai, India

Sep 2021 – Jan 2025

- Developed and tested **statistical models** for insurance **pricing**, **underwriting**, and **fraud risk** using **R**, achieving a **15% improvement** in prediction accuracy for a key fraud detection model.
- Consulted with **customer sales teams** to provide complex analytical results and evidence the value of data-driven risk assessment products, translating technical findings into clear, business-oriented messaging.
- Researched and integrated new **data mining** and **modelling techniques** into the global data science group's workflow, contributing to the development of advanced analytical capabilities.
- Managed and processed **large datasets** from multiple sources, performing **data matching** and **merging** operations to create comprehensive datasets for analysis and model training.
- Built and maintained **Python scripts** for data ingestion and ETL processes, supporting the creation of analytical datasets for the insurance business units.
- Collaborated with internal partners in **Product**, **Sales**, and **Technology** teams, providing technical support and acting as a subject matter resource for data science applications.
- Containerized analytical Python applications using **Docker** and deployed them to **AWS EC2** instances, ensuring reproducible environments for model development and testing.

SKILLS

Data Science & ML – statistical methods, predictive modelling, data mining, regression analysis, logistic regression, hypothesis testing, R, Python, scikit-learn

Data Analysis & BI – Power BI, Tableau, DAX, Power Query, data modelling, data warehousing, exploratory data analysis, A/B testing, time-series analysis

Programming & Scripting – Python, Java, SQL, Spring Boot, pandas, NumPy, Flask, FastAPI

Cloud & DevOps – AWS (EC2, S3, Lambda, CloudWatch), Docker, Jenkins, Git, Linux

Databases – PostgreSQL, MySQL, Oracle

Enterprise Systems – ERP integration support, change management, incident response, Splunk, Dynatrace

PROJECTS

Insurance Risk Modelling with Python and R

- Developed **predictive models** using **Python (scikit-learn)** and **R** to assess underwriting risk for a portfolio of **~50,000 insurance policies**.
- Engineered **feature pipelines** to extract relevant variables from diverse data sources, including policy details and historical claims data, improving model input quality by **20%**.
- Evaluated model performance using standard metrics such as **AUC** and **F1-score**, presenting findings to the analytics team to inform model selection and tuning.

Customer Data Integration for Analytics Platform

- Designed and implemented a **data integration** process using **Python** and **SQL** to merge customer data from multiple legacy systems into a centralized data warehouse.
- Created **ETL scripts** that processed and transformed approximately **5 GB** of data daily, ensuring data consistency and availability for downstream analytical reporting.
- Collaborated with business analysts to define data requirements and validate the integrated dataset, supporting the delivery of accurate operational dashboards.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin

Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India

2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)